

Physics Intro & Kinematics

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- Quantities
- Units
- Vectors
- Displacement
- Velocity
- Acceleration
- Kinematics
- Graphing Motion in one dimension

Some Physics Quantities

Vector - quantity with both magnitude (size) and direction

Scalar - quantity with magnitude only

Vectors:

- Displacement
- Velocity
- Acceleration
- Momentum
- Force

Scalars:

- Distance
- Speed
- Time
- Mass
- Energy

Mass vs. Weight

Mass

- Scalar (no direction)
- Measures the amount of matter in an object

Weight

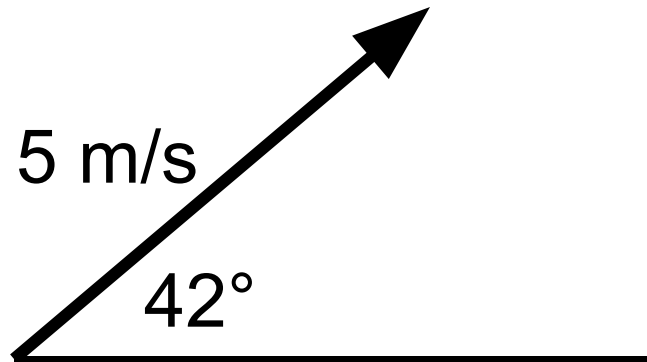
- Vector (points toward center of Earth)
- Force of gravity on an object

On the moon, your mass would be the same, but the magnitude of your weight would be less.

Vectors

Vectors are represented with arrows

- The length of the arrow represents the magnitude (how far, how fast, how strong, etc, depending on the type of vector).
- The arrow points in the direction of the force, motion, displacement, etc. It is often specified by an angle.



Units

Units are not the same as quantities!

Quantity . . . Unit (symbol)

- Displacement & Distance . . . meter (m)
- Time . . . second (s)
- Velocity & Speed . . . (m/s)
- Acceleration . . . (m/s²)
- Mass . . . kilogram (kg)
- Weight & Force . . . newton (N)

SI Prefixes

Little Guys

pico	p	10^{-12}
nano	n	10^{-9}
micro	μ	10^{-6}
milli	m	10^{-3}
centi	c	10^{-2}

Big Guys

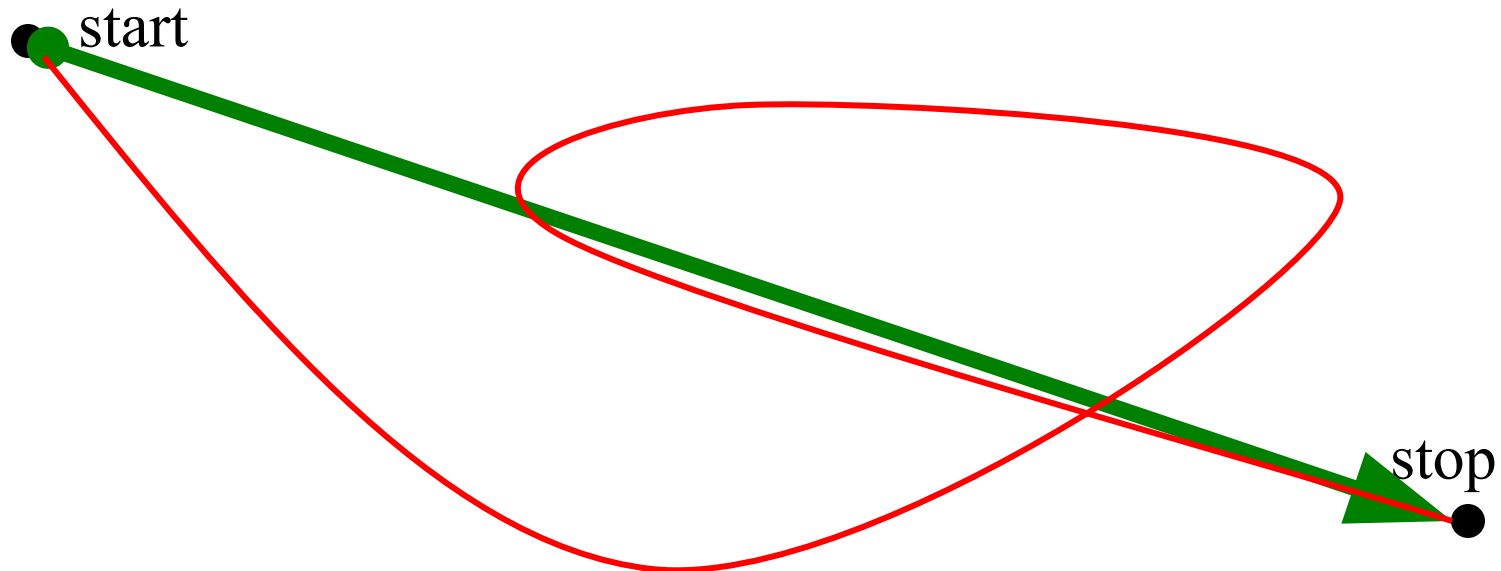
kilo	k	10^3
mega	M	10^6
giga	G	10^9
tera	T	10^{12}

Kinematics definitions

- Kinematics – branch of physics; study of motion
- Position ($\mathbf{x}$) – where you are located relative to some defined reference and some defined direction as positive
- Distance (d) – how far you have traveled, regardless of direction
- Displacement ($\Delta\mathbf{x}$) – change in position (where you are in relation to where you started)

Distance vs. Displacement

- You drive along the path shown in red, and your odometer goes up by 8 miles (your distance, a scalar).
- Your displacement is less than 8 miles. It is the “directed distance” from start to stop (green arrow). This is a vector.
- What if you drove in a full circle? Then your starting and stopping points would be the same. Your distance would be the circle’s circumference; your displacement would be zero!



Speed, Velocity, & Acceleration

- Speed (v) – how fast you go
- Velocity ($\mathbf{v}$) – how fast and which way;
the rate at which position changes
- Average speed (v_{avg} or $\bar{v}$) – distance / time
- Acceleration ($\mathbf{a}$) – how rapidly you speed up,
slow down, or change direction;
the rate at which velocity changes

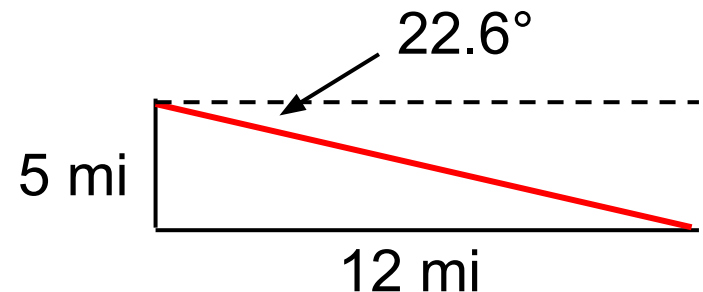
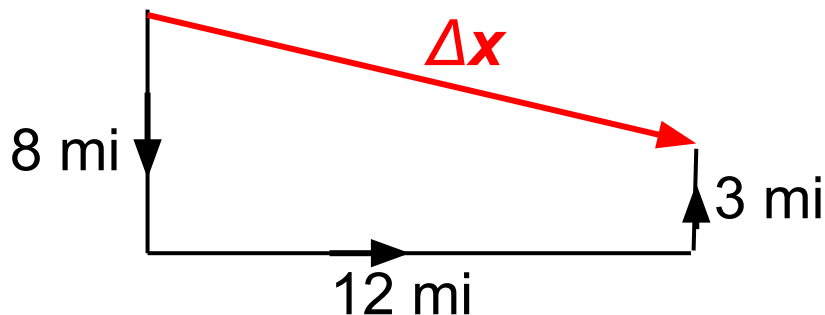
Speed vs. Velocity

- Speed is a scalar (how fast something is moving regardless of its direction). Ex: $v = 20$ mph
- Speed is the magnitude of velocity.
- Velocity is a combination of speed and direction.
Ex: $\mathbf{v} = 20$ mph at 15° south of west
- The symbol for speed is v .
- The symbol for velocity is typewritten in bold: $\mathbf{v}$ or hand written with an arrow: $\vec{v}$
- In general, vectors are written in bold or with an arrow, except sometimes in 1D motion, since a $+/-$ sign can be used to indicate direction. Magnitudes of vectors are written “normally.”

Example

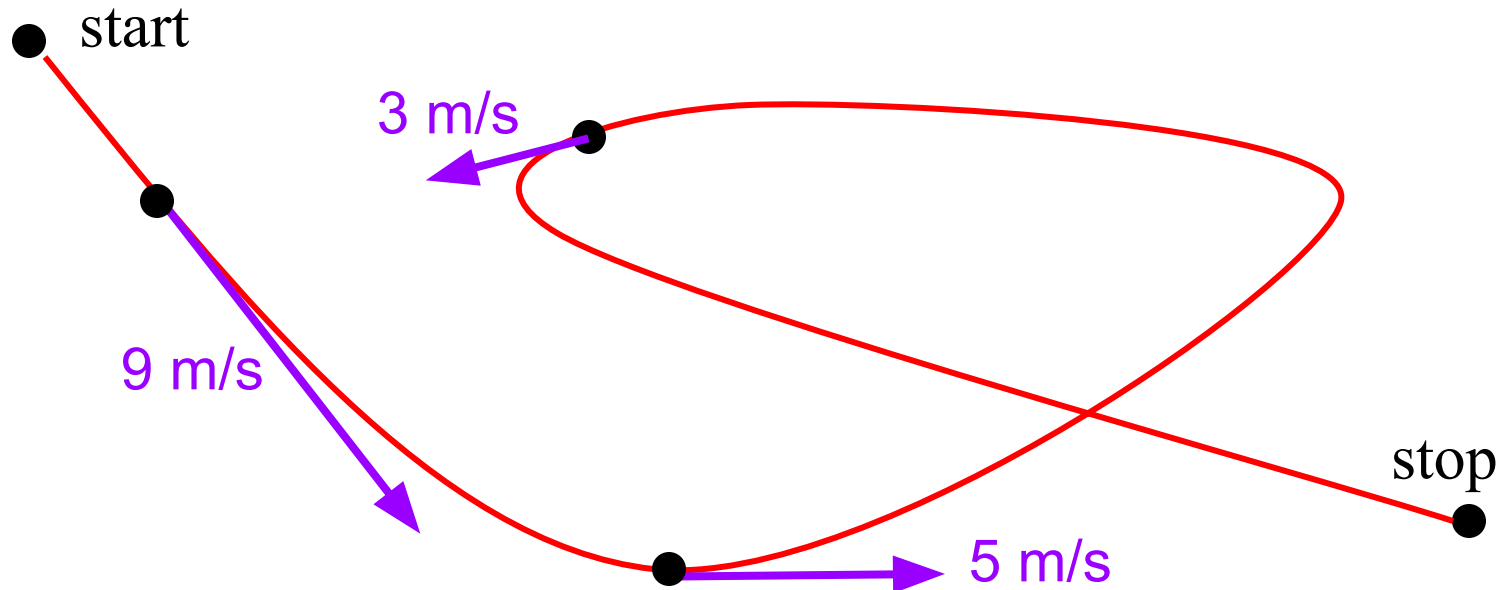
Suppose you drive 8 mi south, 12 mi east, then 3 mi north, and it takes you 30 min. As you drive, your speedometer displays your instantaneous speed, which varies throughout the trip. Find distance, displacement, avg. speed, and avg. velocity.

Answer: $d = 23$ mi. (just add). Using the Pythagorean theorem, $\Delta \mathbf{x} = 13$ mi in the SE direction. Specifically, that direction is about 22.6° south of east, since $\tan^{-1}(5/12) = 22.6^\circ$. Average speed = $23 \text{ mi} / 0.5 \text{ h} = 46 \text{ mph}$. Average velocity = $13 \text{ mi} / 0.5 \text{ h} = 26 \text{ mph}$ at 22.6° south of east.

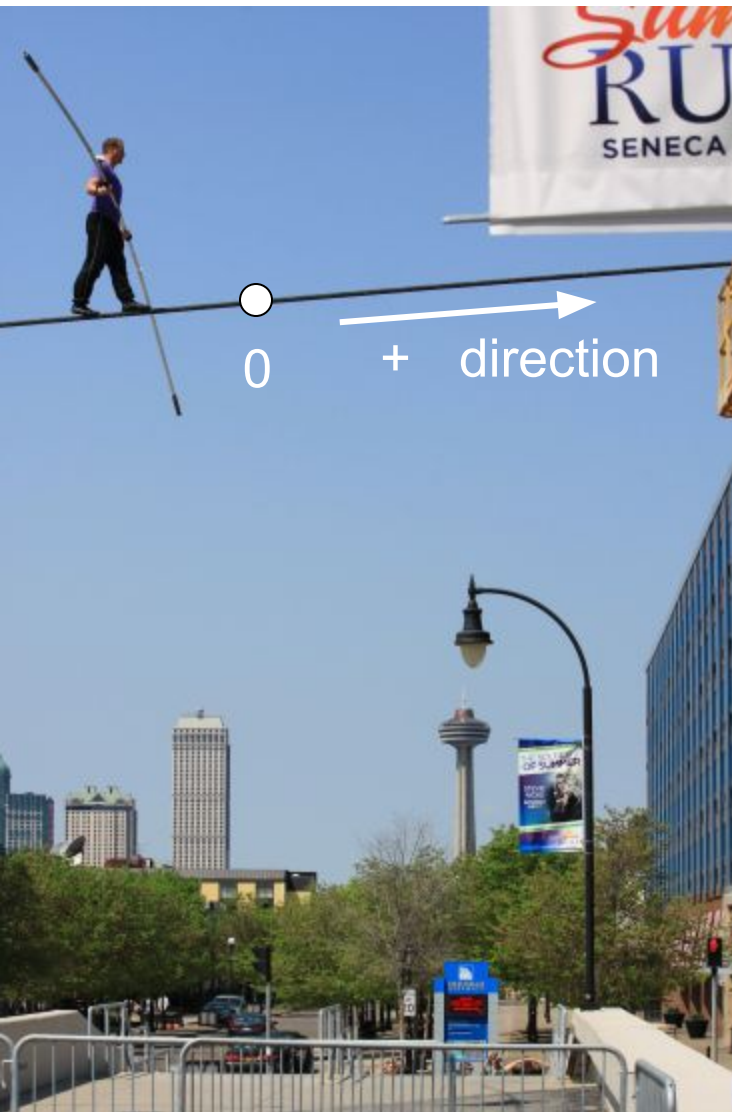


Drawing Velocity Vectors

At any point in time, your velocity vector points tangent to your path in the direction you're heading. Even if you're continually turning, your velocity vector at any moment is "straight ahead"--the direction in which your headlights point. You can draw velocity vectors as long as you like, provided their length is proportional to the speed, as shown. Note: These are instantaneous velocities. The avg. velocity vector is parallel to the displacement vector.



Velocity in 1D



A tightrope walker can only move forwards and backwards--one dimensional motion. An arbitrary point on the rope can be called zero, and an arbitrary direction can be called positive. Now the rope is essentially a number line. In the picture, Wallenda's position is negative, but (assuming he's walking forward) his velocity is positive. If something is moving in the positive direction, its velocity is positive, regardless of where its located. If Wallenda were on the other side of the origin and walking backwards, his position would be positive and his velocity negative. Thus the sign of velocity is easy. It's simply which way something is moving!

Acceleration

Acceleration is how fast you speed up, slow down, or change direction; it's the rate at which velocity changes. Each table is velocity vs. time for a car moving in the positive direction.

t (s)	v (mph)
0	55
1	57
2	59
3	61

$$a = +2 \text{ mph/s}$$

(velocity increasing by 2 mph
each second)

t (s)	v (m/s)
0	34
1	31
2	28
3	25

$$a = -3 \frac{\text{m/s}}{\text{s}} = -3 \text{ m/s}^2$$

(velocity decreasing by 3 m/s
each second)

Velocity & Acceleration Sign Chart

		<i>VELOCITY</i>	
<i>ACCELERATION</i>		+	-
	+	Moving forward; Speeding up	Moving backward; Slowing down
	-	Moving forward; Slowing down	Moving backward; Speeding up

Acceleration Clarification

In common English the word *accelerate* means to speed up, and *decelerate* means to slow down. In physics, to accelerate means to speed up, slow down, or turn, i.e., to change to the velocity vector in any way. When moving in the + direction, deceleration is synonymous with negative acceleration. But negative acceleration does not always mean slowing down! It means decreasing velocity, either + velocity becoming less +, or - velocity becoming more -. Examples:

Let's say north is +. Moving north at 6 m/s and slowing down 4 m/s is negative acceleration, since $+6 \rightarrow +4$ is a decrease.

Moving south at 6 m/s and slowing down 4 m/s is positive acceleration, since $-6 \rightarrow -4$ is an increase! *Tricky.*

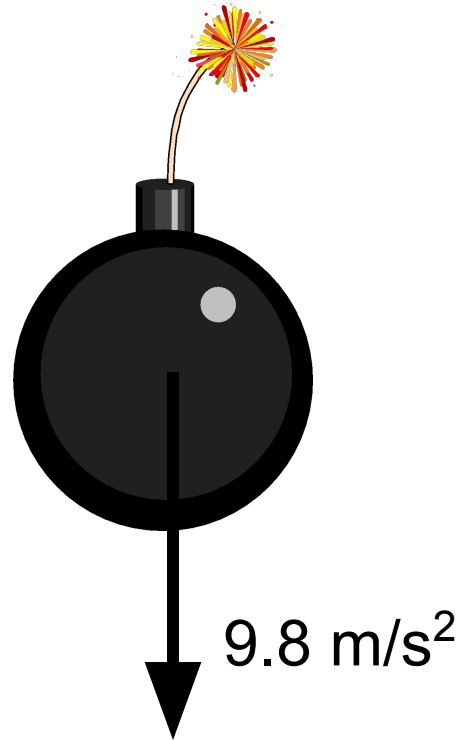
Moving north at 4 m/s and speeding up to 6 m/s is positive acceleration, since $+4 \rightarrow +6$ is an increase.

Moving south at 4 m/s and speeding up to 6 m/s is negative acceleration, since $-4 \rightarrow -6$ is a decrease! *Tricky.*

Acceleration due to Gravity

Near the surface of the Earth, all objects accelerate at the same rate (ignoring air resistance).

$$a = -g = -9.8 \text{ m/s}^2$$



This acceleration vector is the same on the way up, at the top, and on the way down!

Interpretation: The negative sign is because we're choosing up as + and gravity always accelerates things downward. Velocity *decreases* by 9.8 m/s each second, meaning velocity is becoming less positive (on the way up) or more negative (on the way down). Even at its highest point (where velocity is zero for an instant) acceleration is still -g, since velocity is in the process of changing there.

Kinematics Quantities & Equations

t ... time

d ... distance

Δx ... displacement

v_0 ... initial velocity

v_f ... final velocity

Δv ... change in velocity, $v_f - v_0$

a ... acceleration

$$d = v_{avg} t, \text{ for } v_{avg} = \text{avg. speed} \quad \Delta x = v_{avg} t, \text{ for } v_{avg} = \text{avg. velocity}$$

When acceleration is uniform:

$$v_f = v_0 + a t \quad \Delta x = v_0 t + \frac{1}{2} a t^2 \quad v_f^2 - v_0^2 = 2a \Delta x$$

$$v_{avg} = \frac{1}{2} (v_0 + v_f), \text{ for } v_{avg} = \text{avg. velocity}$$

Kinematics Derivations

$$a = \Delta v / \Delta t \quad (\text{by definition})$$

$$a = (v_f - v_0) / t$$

$$\Rightarrow v_f = v_0 + at$$

$v_{avg} = (v_0 + v_f) / 2$ will be proven when we do graphing.

$$\Delta x = v_{avg} t = \frac{1}{2} (v_0 + v_f) t = \frac{1}{2} (v_0 + v_0 + at) t$$

$$\Rightarrow \Delta x = v_0 t + \frac{1}{2} at^2$$

(cont.)

Kinematics Derivations (cont.)

$$v_f = v_0 + at \quad \Rightarrow \quad t = (v_f - v_0)/a$$

$$\Delta x = v_0 t + \frac{1}{2} a t^2$$

$$\Rightarrow \Delta x = v_0 [(v_f - v_0)/a] + \frac{1}{2} a [(v_f - v_0)/a]^2$$

$$\Rightarrow v_f^2 - v_0^2 = 2a\Delta x$$

Note that the top equation is solved for t and that expression for t is substituted twice (in red) into the Δx equation. You should work out the algebra to prove the final result on the last line.

Diagrams & Units, & Rounding

Diagrams with the given question mark quantities are *required* and worth points on quizzes. They are helpful, especially with more complicated problems.

Often it's easier to solve for the needed quantity before plugging in numbers. That is, algebra first; plug in second.

Correct units are required on your answer. Technically, the quantities should be plugged in with their units:

$a = (v_f^2 - v_o^2) / (2 \Delta x) = [(25 \text{ m/s})^2 - 0^2] / (2 * 21 \text{ m})$
 $= 14.9 \text{ m/s}^2$. Note that $(\text{m/s})^2 / \text{m} = \text{m/s}^2$. However, I will only require that have the correct units at the end. Be careful that your units are compatible, though. Often you will have to convert before plugging in!

Rounding

In Sample #1, we had $(25^2 - 0^2) / (2 * 21)$, which is 14.8809524.... The more precise the better, right? NO! Do **not** write your final answer with a load of decimal places unless all of your givens were like that.

Technically, you should follow the rules of significant figures. Among the givens, the fewest number of sig figs is 2, so we should write the answer as 15 m/s².

I'm not super picky. 14.9 m/s² is fine, but 14.88095 would lose points for two reasons...not rounding and no units.

Why Rounding Matters

Let's say you walk 2 m in 3 s. What's your average speed? Saying 0.66666667 m/s is nutty. Here's why.

In math $2 = 2.0 = 2.00 = \dots$, but in science 2 m means 2 m give or take around a meter. No measurement is infinitely precise. Uncertainty occurs in the last decimal place. Likewise, 3 s means 3 s give or take a second or so. However, 0.66666667 m/s means you have no uncertainty about the speed until the hundred millionths of a m/s! There is way so much original uncertainty can yield an answer with so little uncertainty.

Moral of the story: Don't go nuts with the decimal places!

Sample Problem #2

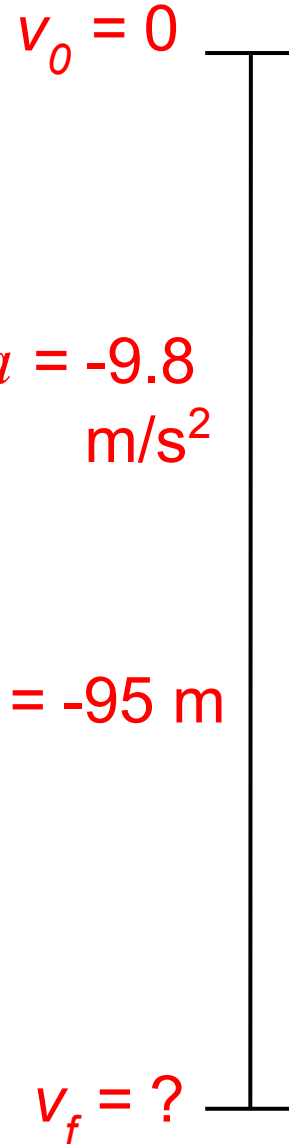
A banjo is dropped from 95 m up. Find its (a) impact velocity and (b) falling time. Ignore air resistance.

Solution: (a) “Dropped” is code for $v_0 = 0$ (unless dropped while moving). $a = -g$, since it’s falling on earth. Δx is negative since banjos fall down, not up. We seek the velocity just as it’s impacting the ground, v_f .

$v_f^2 - v_0^2 = 2a \Delta x$ again has all the quantities in our picture.

$$\begin{aligned} v_f &= \pm[v_0^2 + 2a \Delta x]^{1/2} = \pm[0^2 + 2(-9.8)(-95)]^{1/2} \\ &= \pm 43.1509 \approx -43 \text{ m/s.} \end{aligned}$$

(We choose the negative root since the banjo is moving in the negative direction.)



Sample #2 (cont.)

(b) The falling time (hang time) picture is slightly different. The equation that relates these quantities is $\Delta x = v_0 t + \frac{1}{2} a t^2$. So, we must solve the quadratic $-95 = \frac{1}{2}(-9.8)t^2$. This yields $t = 4.4$ s.

Alternatively, we could have used the v_f we found in part (a) to find t . The equation that works in this case is $v_f = v_0 + a t$.

So $t = (v_f - v_0) / a = (-43.1509 - 0) / (-9.8) = 4.4$ s...same answer as above. Note that the unrounded version of v_f was used. Don't round intermediate answers! The disadvantage of the second method is that an error in part (a) would cause an error in (b).

$$v_0 = 0$$

$$a = -9.8 \text{ m/s}^2$$

$$\Delta x = -95 \text{ m}$$

$$t = ?$$

Sample Problem #3

A jack o'lantern is shot straight up from a pit 12.5 m below ground at 38 m/s. (a) Find its max height above ground.

Solution: We will generally ignore air resistance unless told otherwise. This time our experiment begins low and end high. $v_o > 0$, since it's fired upward; and $v_f = 0$ at the high point, since it must stop for an instant before turning around. Displacement is our unknown. We'll find it first, then adjust for the pit depth.

$$v_f^2 - v_o^2 = 2a \Delta x, \quad \text{so} \quad \Delta x = (v_f^2 - v_o^2) / 2a$$

$= (0^2 - 38^2) / (2 * -9.8) = 73.6735 \text{ m}$. Subtracting off 12.5 m, since it began below ground, and rounding gives us a max height of about 62 m.

$$v_f = 0$$

$$a = -9.8 \text{ m/s}^2$$

$$\Delta x = ?$$

$$v_o = 38 \text{ m/s}$$

Sample #3 (cont.)

(b) At what times is the jack o'lantern at ground level?

Solution: Now the experiment begins below ground and ends at ground level. It is the displacement that matters, not the distance travelled! It ends up 12 m above where it began. There are two times when it's at ground level, once on the way up, and then again on the way back down. The math will give us both of these times:

$$\Delta x = v_0 t + \frac{1}{2} a t^2 \Rightarrow 12 = 38t + \frac{1}{2} (-9.8)t^2.$$

$\Rightarrow$ means “implies,” not “equals.” Solving the quadratic for t , we get 0.33 s (while ascending) and 7.4 s (while descending).

Practice: Find its impact speed on the ground (12 m up).

Answer: about 35 m/s ($v_f < 0$, but *speed* > 0).

$$a = -9.8 \text{ m/s}^2$$

$$t = ?$$

$$\Delta x = 12 \text{ m}$$

$$v_0 = 38 \text{ m/s}$$



Solving Equations via Graphing

Solving quadratic equations with quadratic formula is a drag, especially if there are decimals involved. You're best off not doing that.

Instead, use technology! On homework you can simply type the equation directly into Wolfram Alpha. On quizzes you should use a graphing calculator. There are two ways to do it. One is by graphing.

Get everything over to one side. In the last problem we had $12 = 38t + \frac{1}{2}(-9.8)t^2$. You'd enter $y = -12 + 38x + 0.5(-9.8)x^2$. These equations obviously have the same solutions, but in the second one you can look for x-intercepts on the graph. Sometimes this requires modifying your window.

Solving Equations with the TI Solver

To solve $12 = 38t + \frac{1}{2}(-9.8)t^2$, on your TI: MATH button $\rightarrow$ Solver $\rightarrow$ eqn: $0 = -12 + 38x + 0.5 * -9.8x^2$ $\rightarrow$ ENTER $\rightarrow$ $x = 100$ $\rightarrow$ ALPHA ENTER.

When you enter 100 for x , just overwrite what was there. There is nothing special about 100. This initial guess is just a starting point for the calculator's algorithm. If the equation has more than one zero, it should find the one closer to your initial guess. In this problem, the calculator returns the later time (on the way down). Entering a small or negative number should return the earlier time (on the way up).

The nice thing about graphing or using the solver is that these techniques work with any type of equation. Be sure you understand these methods. The quadratic formula is more error-prone and time-consuming. Trust me. Practice using your calculator!!

Multi-step Problems

(not on summer homework)

1. How fast should you throw a kumquat straight down from 40 m up so that its impact speed would be the same as a mango's dropped from 60 m?

Answer: 19.8 m/s

2. A dune buggy accelerates uniformly at 1.5 m/s^2 from rest to 22 m/s. Then it brakes uniformly until it stops 2.5 s later. Find the total distance traveled.

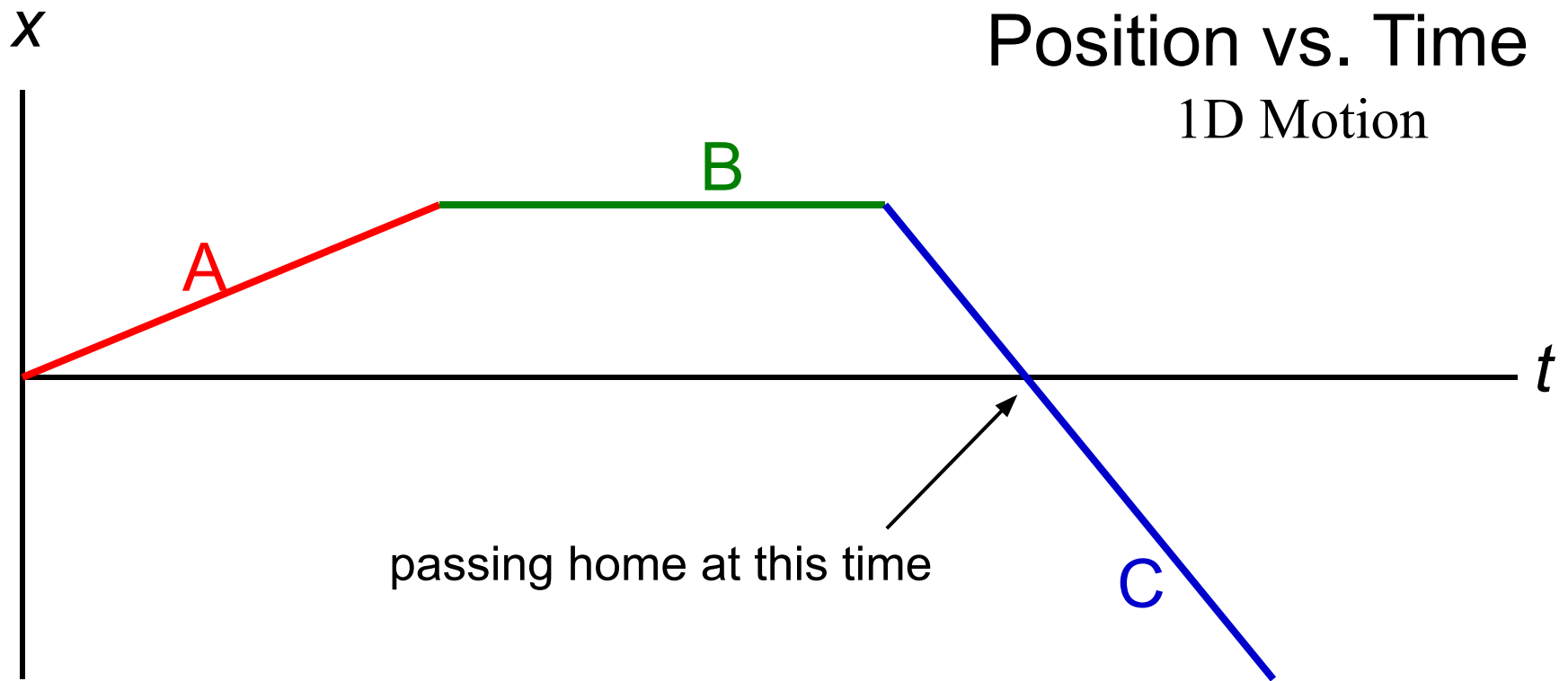
Answer: 188.83 m

Graphing 1D Motion

In this unit we're only considering objects moving along a line. We'll branch out to higher dimensions later. For now we'll learn to graph position vs. time, velocity vs. time, and acceleration vs. time for 1D motion. The graphs will be 2D, but that's only because time is on one axis.

Example: Suppose a remote-controlled car starts at “home,” an arbitrary reference point, and heads north (our arbitrarily chosen + direction) at a constant, slow speed. It then stops and goes nowhere a while. After that it heads back toward home and passes it up at a faster, constant speed.

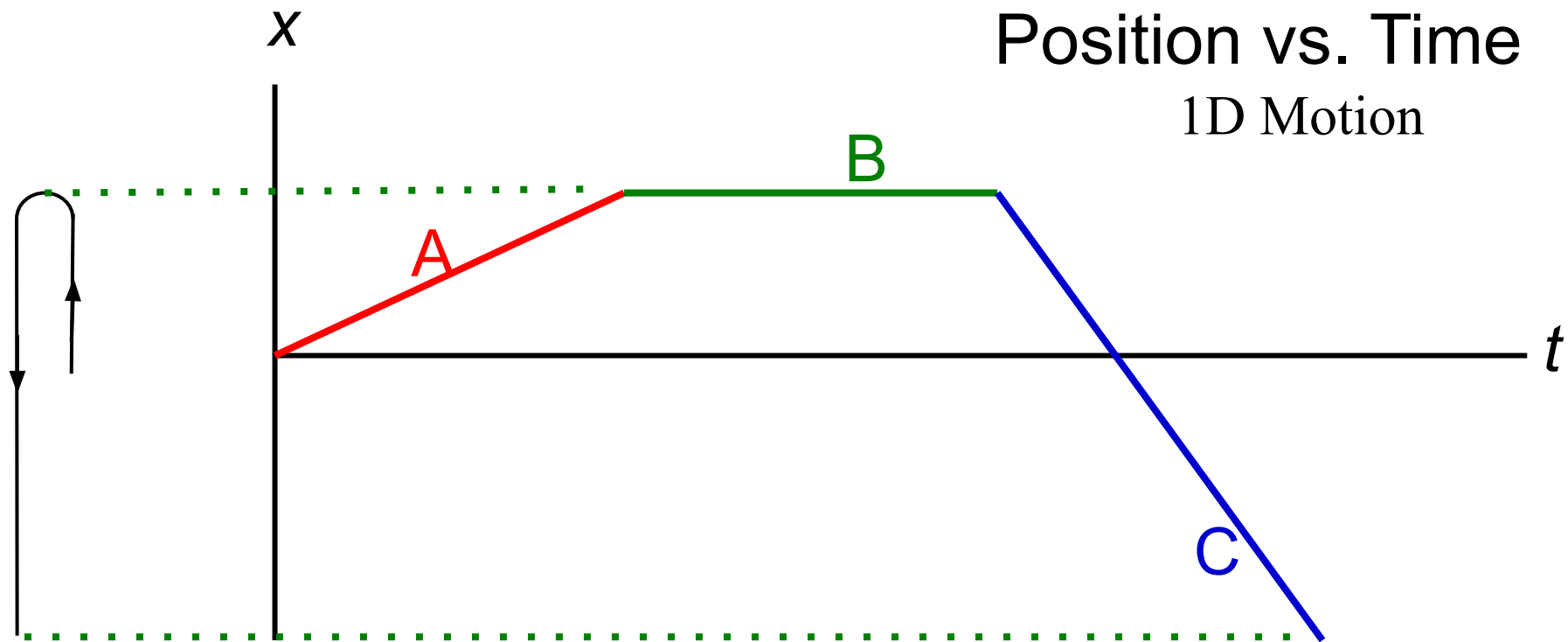
Check out the position vs. time (x vs. t) graph on the next slide. Note: where the car is fast, the slope is steeper; where it's moving N, the slope is positive; where it's moving S, the slope is negative.



A ... Starts at home (origin); goes N slowly, constant speed

B ... Not moving (position remains constant as time progresses)

C ... Turns around and goes in the other direction quickly and at constant speed, passing up home



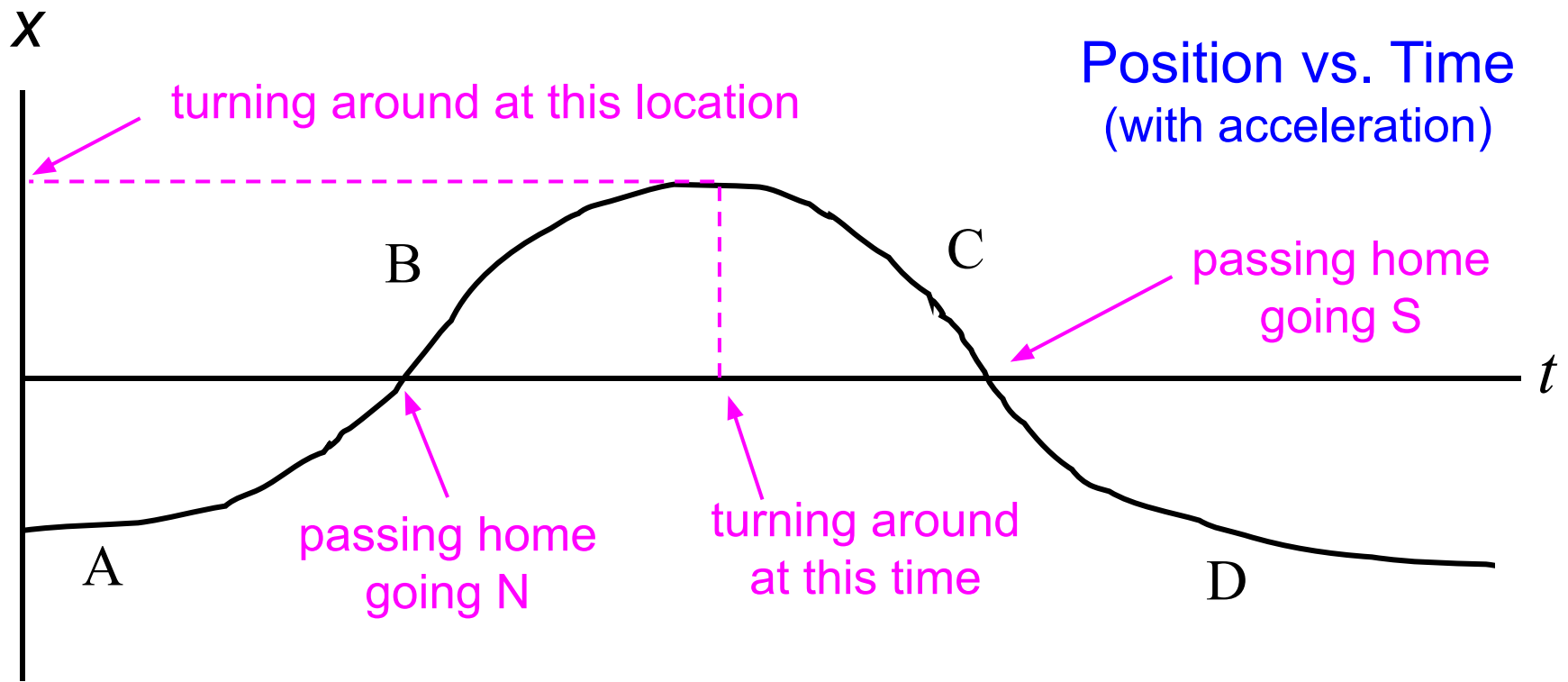
The car's motion is NOT “uphill--plateau--steep downhill”!
The path of the actual motion is shown to the left of the graph. It's purely along a single line--first N, then even more S. This path does not indicate speed, nor does it show the pause, as the graph does. The distance travelled is the total length of this path, whereas the displacement is S and its magnitude is the distance between starting and stopping points. Make your own graphs [here](#).

Graphing with Acceleration

In the last example there was no acceleration, hence all three parts of the graph were linear. The abrupt changes in velocity (where the slope suddenly changes) is unrealistic. In real life the graph would have curves rather than corners between the line segments.

If there is uniform acceleration, the position graph will be a **parabolic** arc, instead of a line segment. On the position graph on the next slide, note that when velocity increases, the slope increases (by becoming more + or less -), and that the graph is concave up. When velocity decreases, the slope decreases (by becoming less + or more -), and the graph is concave down.

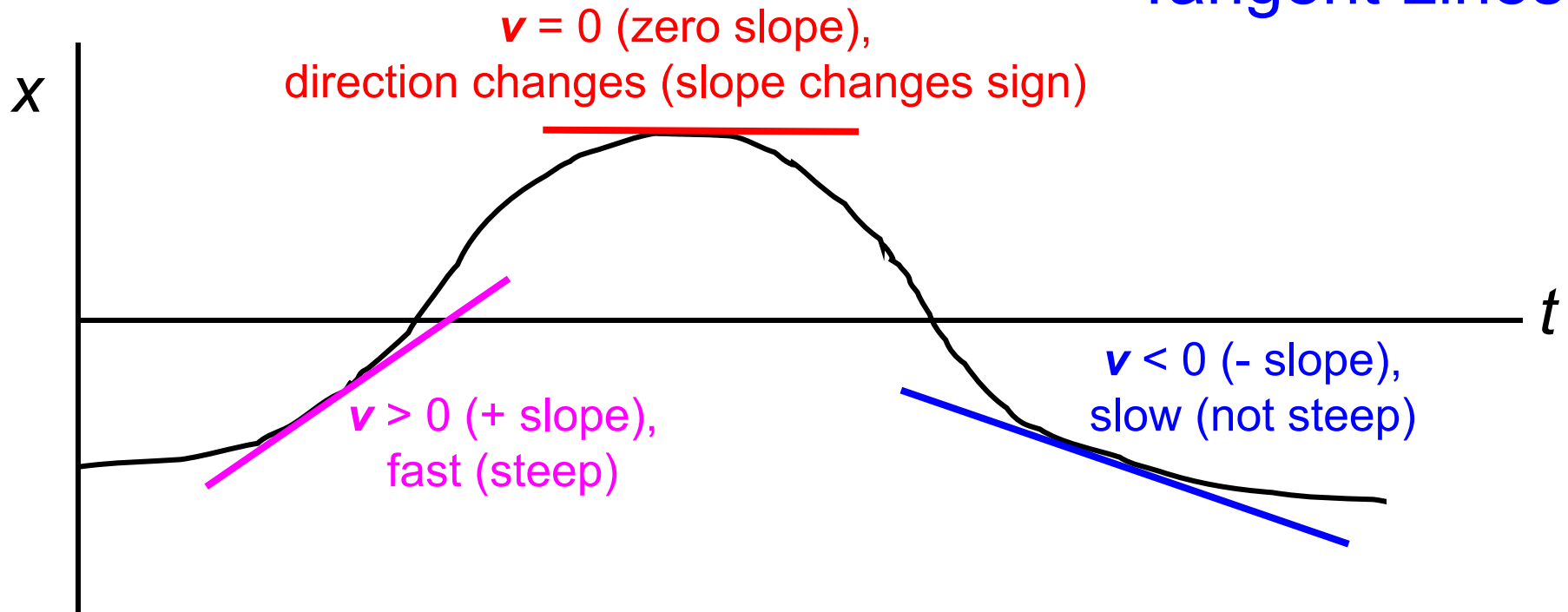
Also, time axis intercepts are times when “home” is passed.



Let's choose north as the + direction again.

- A ... Start from rest S of home; increase speed gradually, heading N
- B ... Pass home; gradually slow to a stop (still moving N)
- C ... Turn around; gradually speed back up again heading S
- D ... Continue heading S; gradually slow to a stop near the starting point

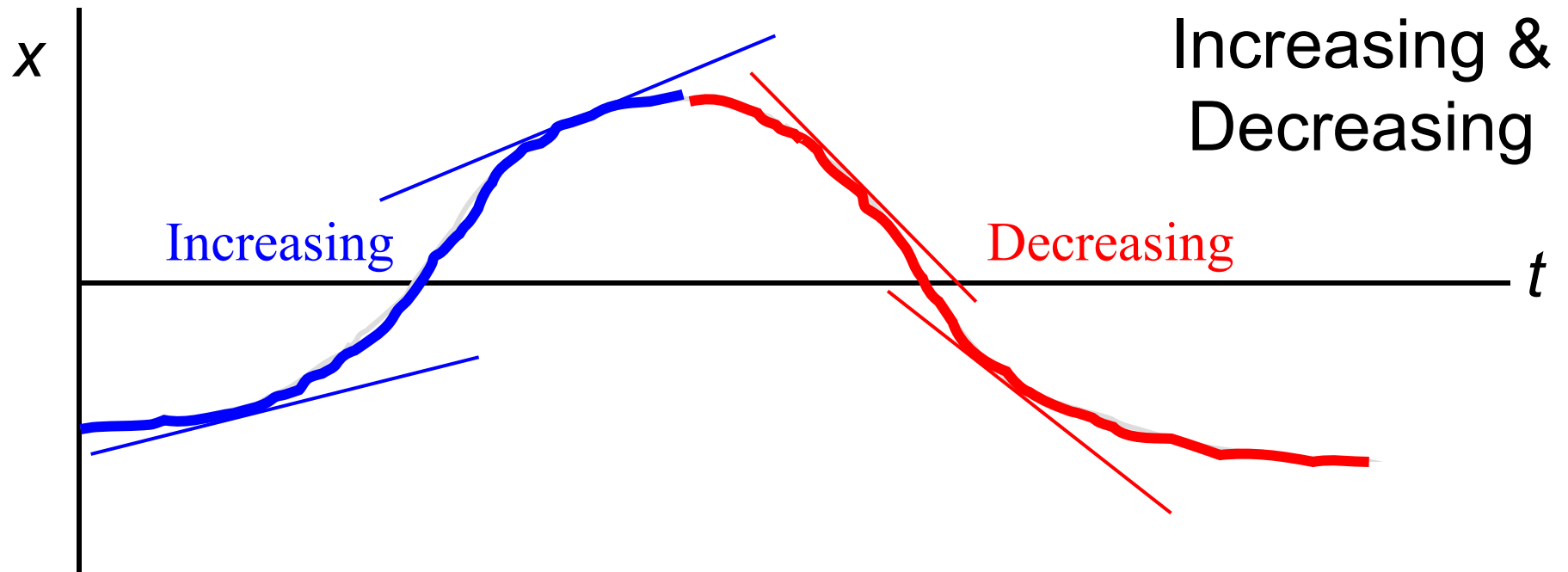
Tangent Lines



On a **position vs. time** graph:

SLOPE	VELOCITY
Positive	Positive
Negative	Negative
Zero	Zero

SLOPE	SPEED
Steep	Fast
Gentle	Slow
Flat	Zero

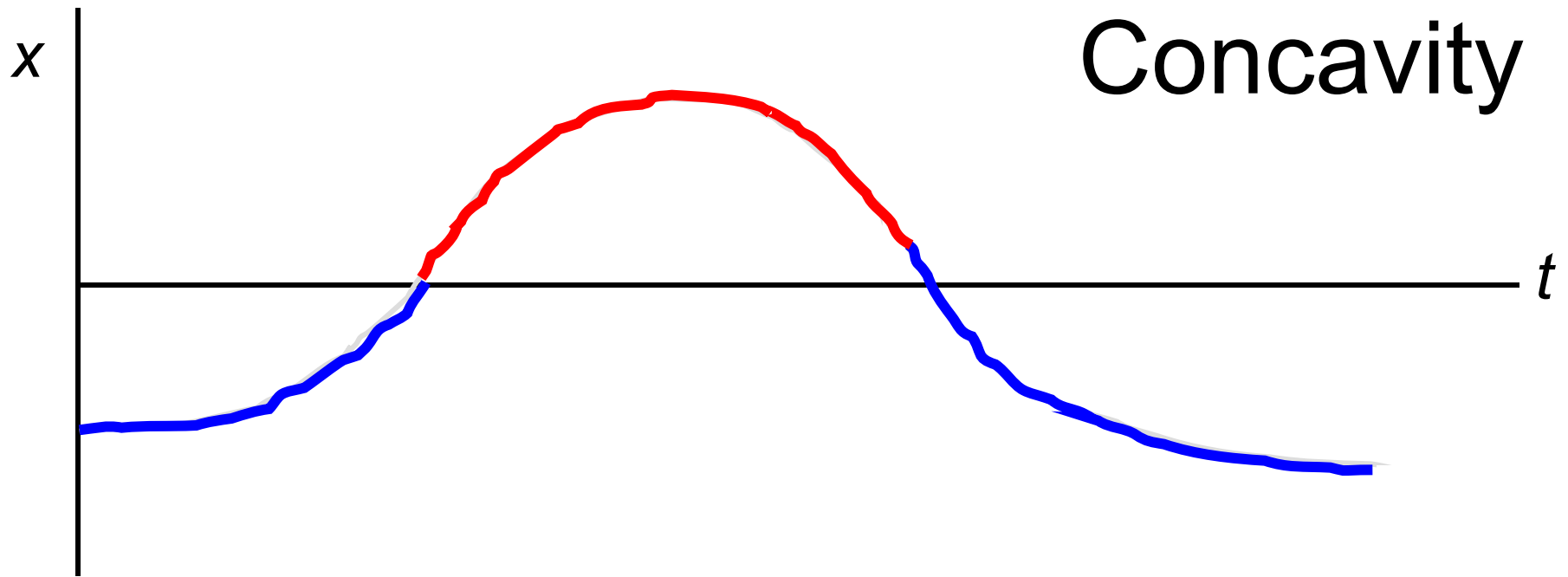


On a **position vs. time** graph:

Increasing means moving in the positive direction
(+ slopes, + velocity).

Decreasing means moving in the negative direction
(- slopes, - velocity).

*Note: Velocity is the slope of the tangent line (with the sign).
Speed is the absolute value of the tangent slope.*



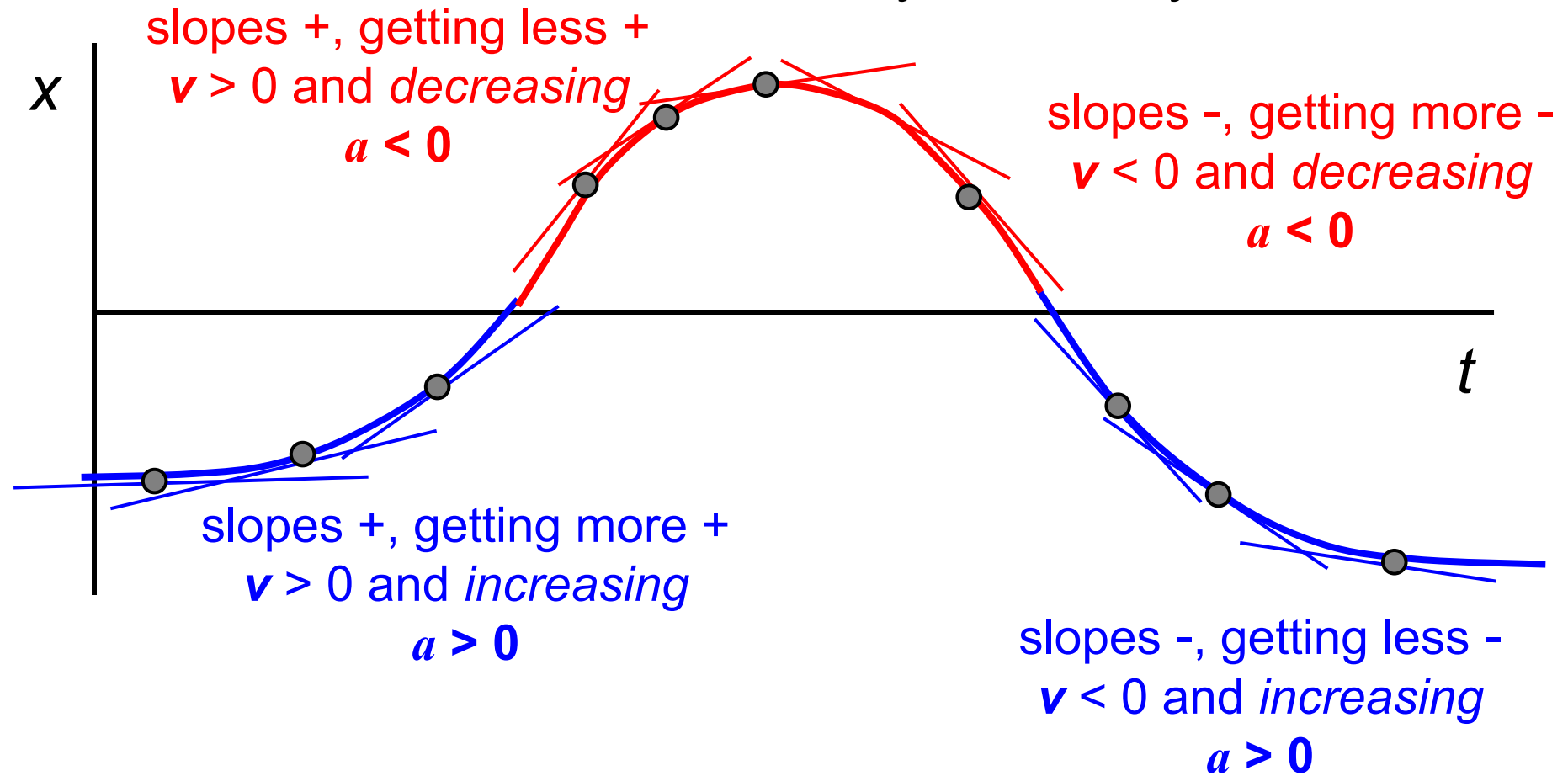
On a position vs. time graph:

Concave up means positive acceleration.

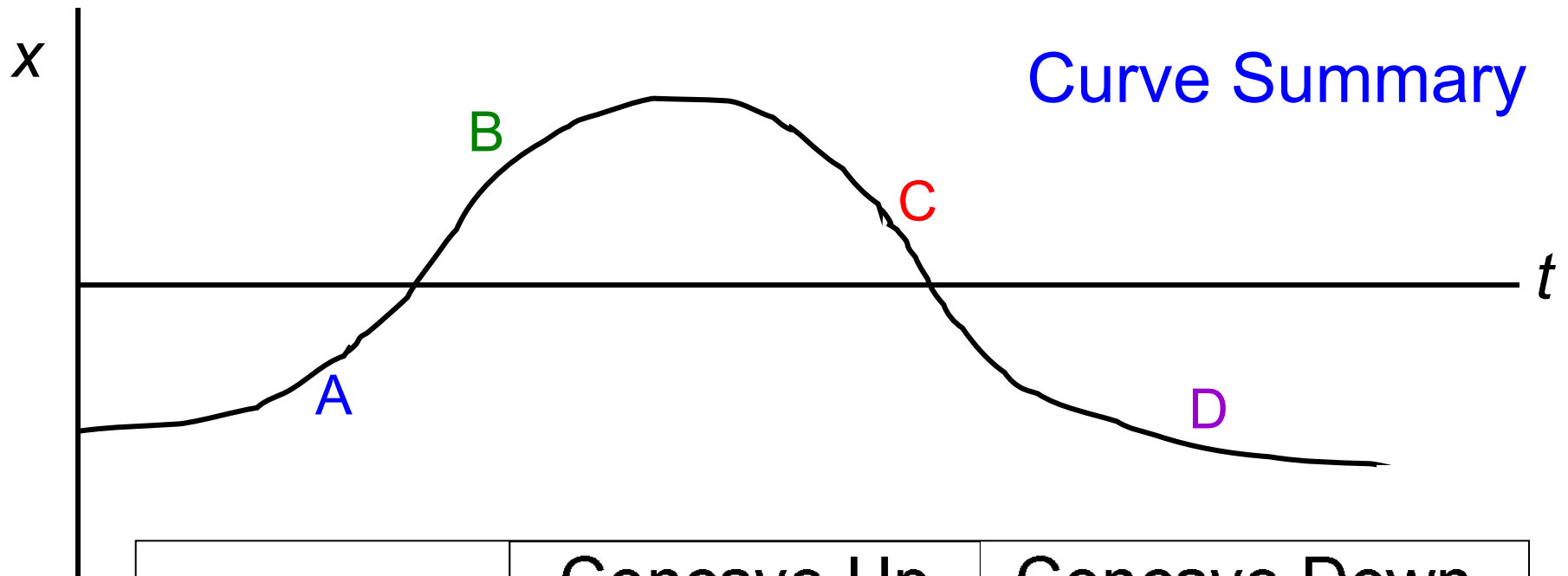
Concave down means negative acceleration.

See why on the next slide.

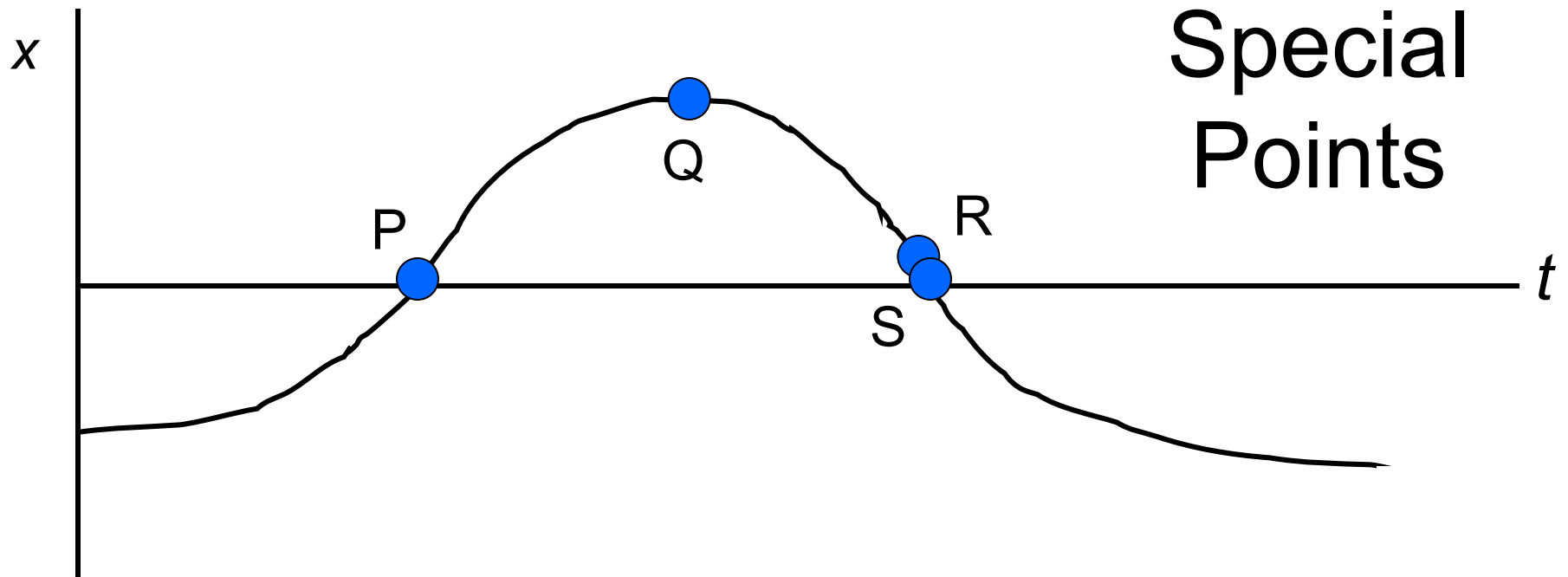
Why Concavity = Acceleration



This is true for x vs. t graphs only! What matters for acceleration is whether velocity (slope) is increasing or decreasing. Velocity increases in the concave up (blue) regions, where tangent slopes becomes more positive or less negative.



	Concave Up	Concave Down
Increasing	$v > 0$ $a > 0$ (A)	$v > 0$ $a < 0$ (B)
Decreasing	$v < 0$ $a > 0$ (D)	$v < 0$ $a < 0$ (C)



Inflection Point	P, R	Change of concavity (acceleration changes sign)
Peak or Valley	Q	Turning point (velocity changes sign)
Time Axis Intercept	P, S	Times when you are at “home”

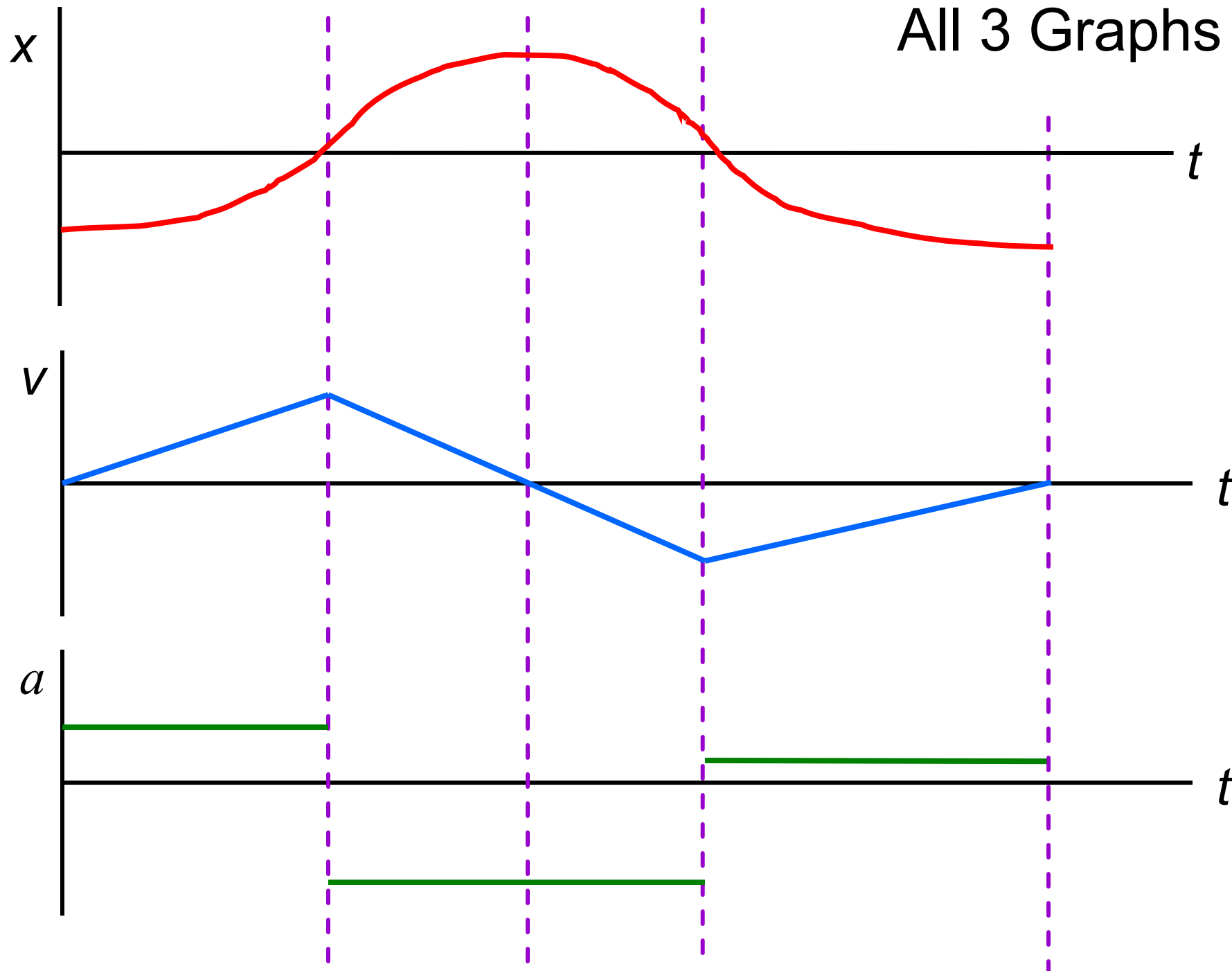
Graphing Velocity and Acceleration vs. Time

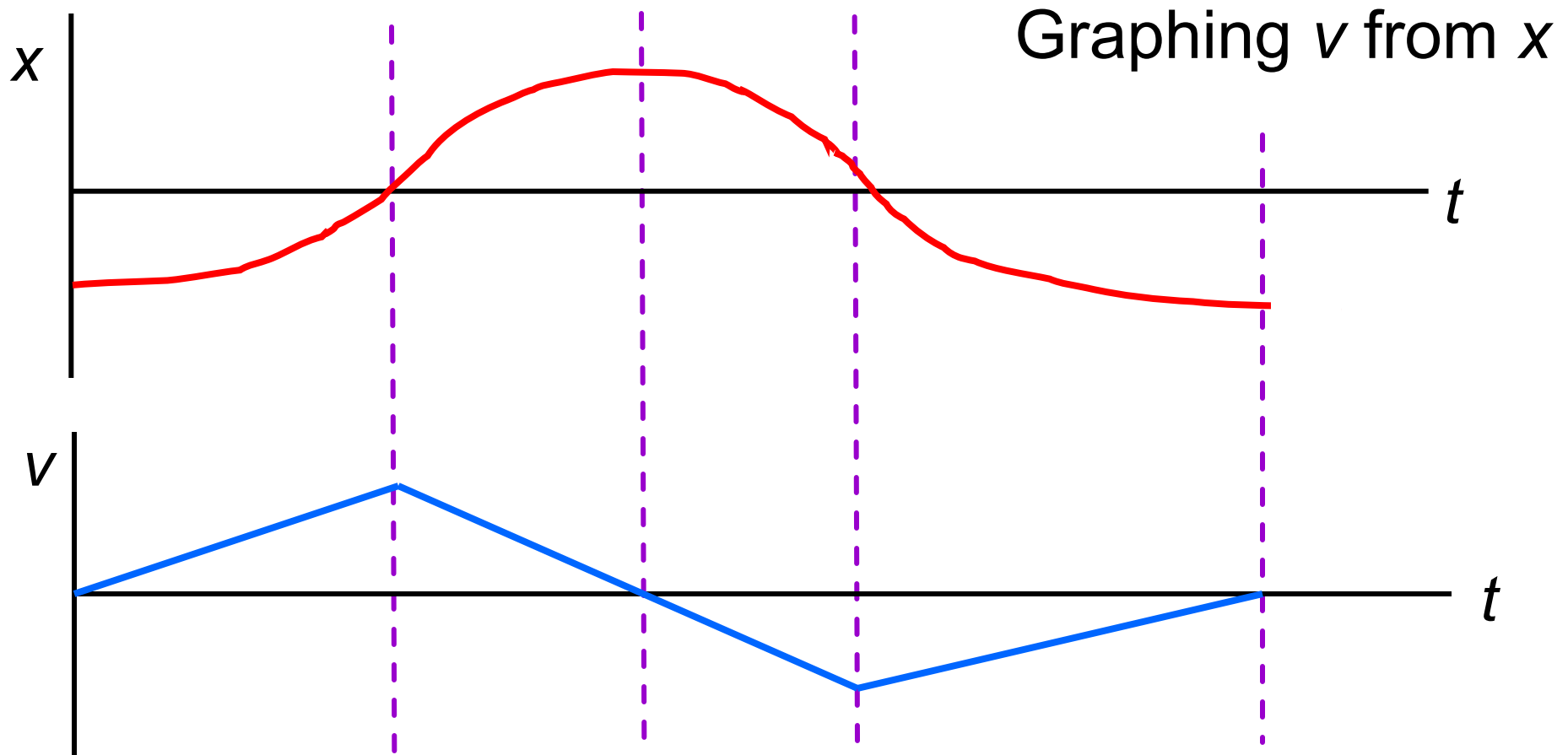
Velocity is the rate at which position changes--the slope of the position graph. Thus in order to create a velocity graph, we must map the slopes of tangent lines on the position graph. Similarly, we can make an acceleration graph by mapping the slopes of the velocity graph.

Note on the next slide how v starts at zero, where the x graph has zero slope. As the slopes of x increase, so do the values of v . At the inflection point, the slope is at a max, so the v graph peaks. The slopes of the x graph then decline, as do the v values. When x peaks, the slope is zero, and so is v . Then the slopes of the x graph becomes negative, along with velocity.

At any point on the position graph, the slope at that time is the velocity at that time. Likewise, at any point on the velocity graph, the slope at that time is the acceleration at that time.

All 3 Graphs

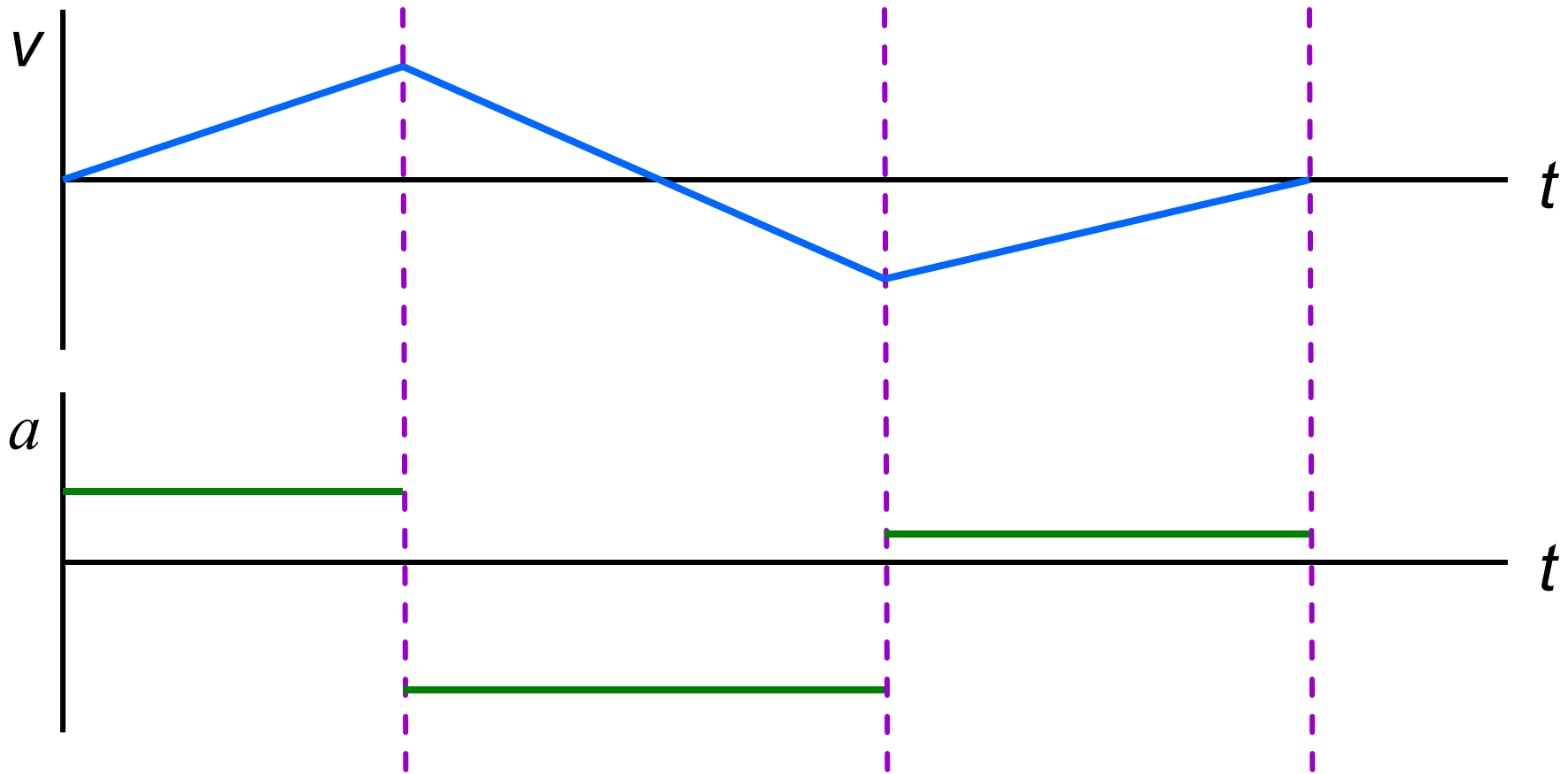




- Line up the graphs vertically.
- Draw vertical dashed lines at turning points and inflection points.
- Map the slopes of the position graph onto the velocity graph.
- A red peak or valley means a blue time intercept. (0 **slope** = 0 **vel.**)

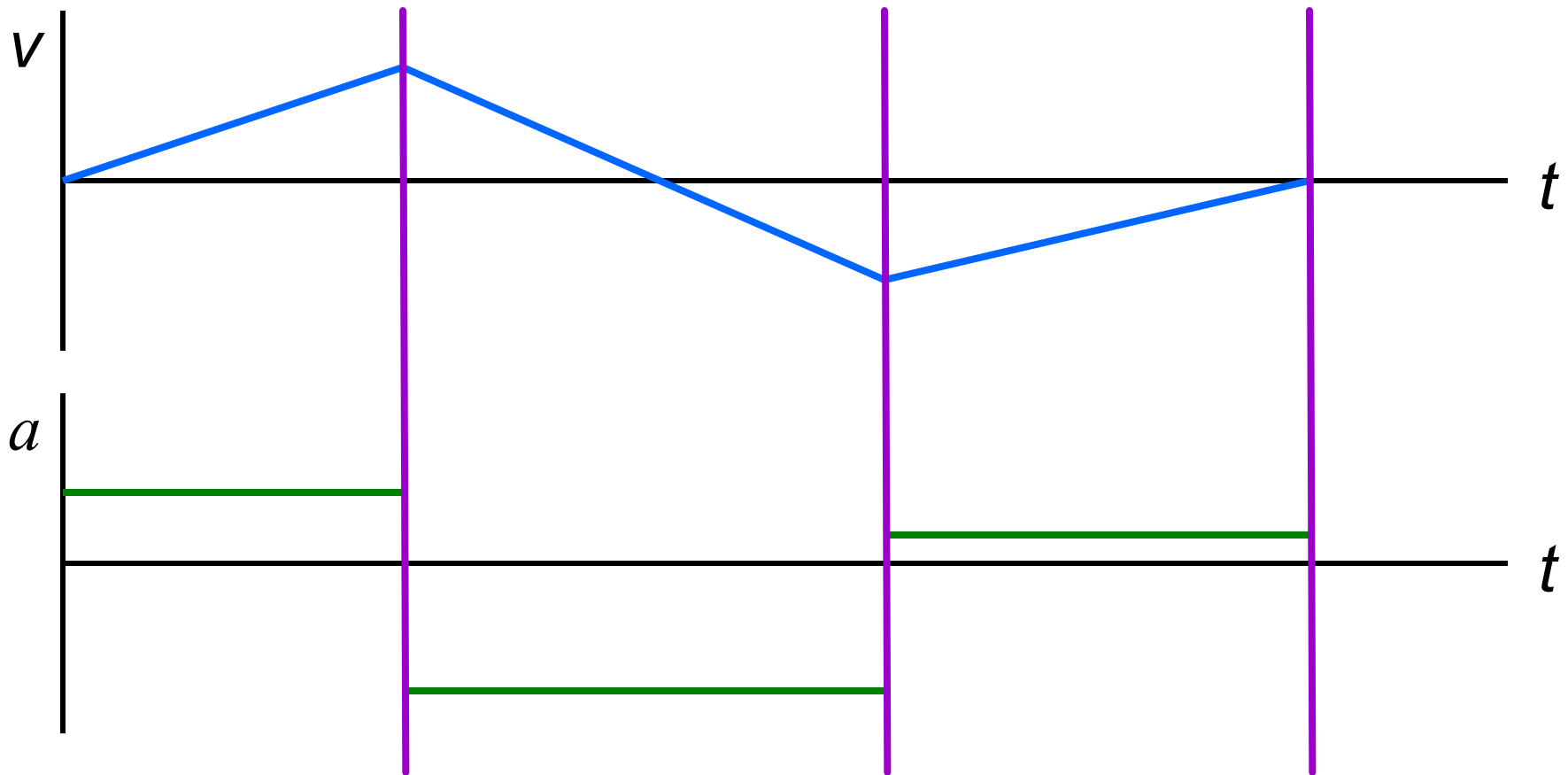
Graphing a from v

The same rules apply in making an acceleration graph from a velocity graph. Just graph the slopes! Note: a positive constant slope on the velocity (blue) graph means a positive constant acceleration (green). The steeper the blue slope, the farther the green segment is from the time axis.

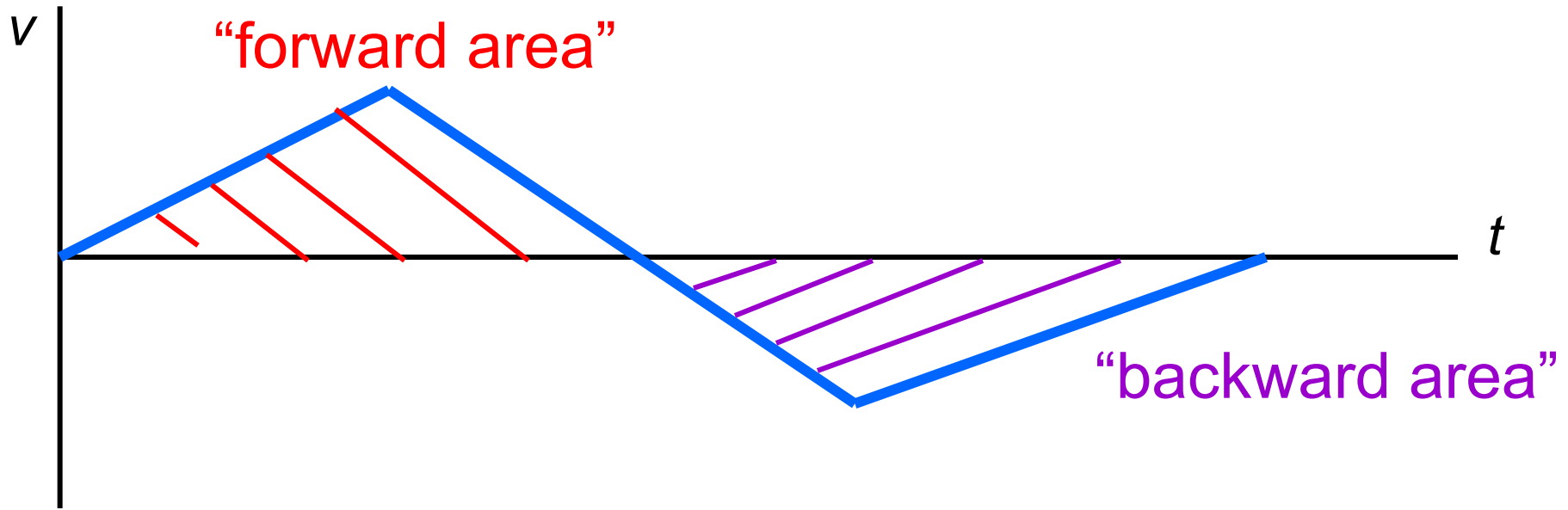


Real life

Note how the v graph is pointy and the a graph is discontinuous. In real life, the blue corners would be smooth curves and the green segments would be connected. In our class, however, we'll mainly deal with constant acceleration.



Area under a velocity graph

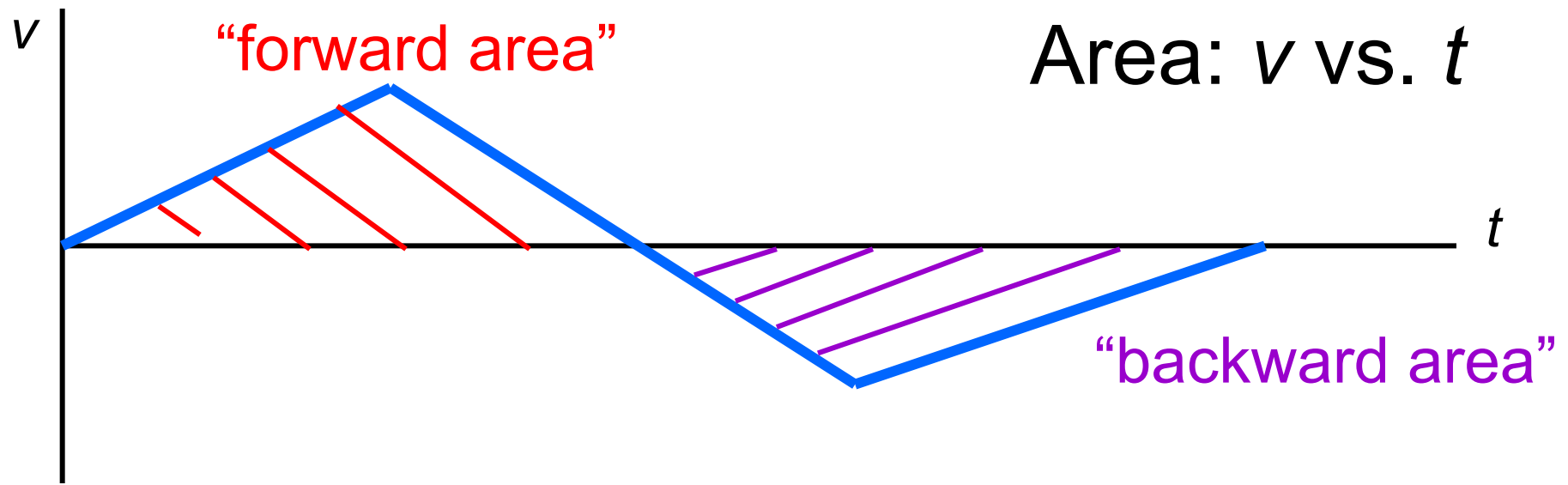


Area above the time axis = forward (positive) displacement.

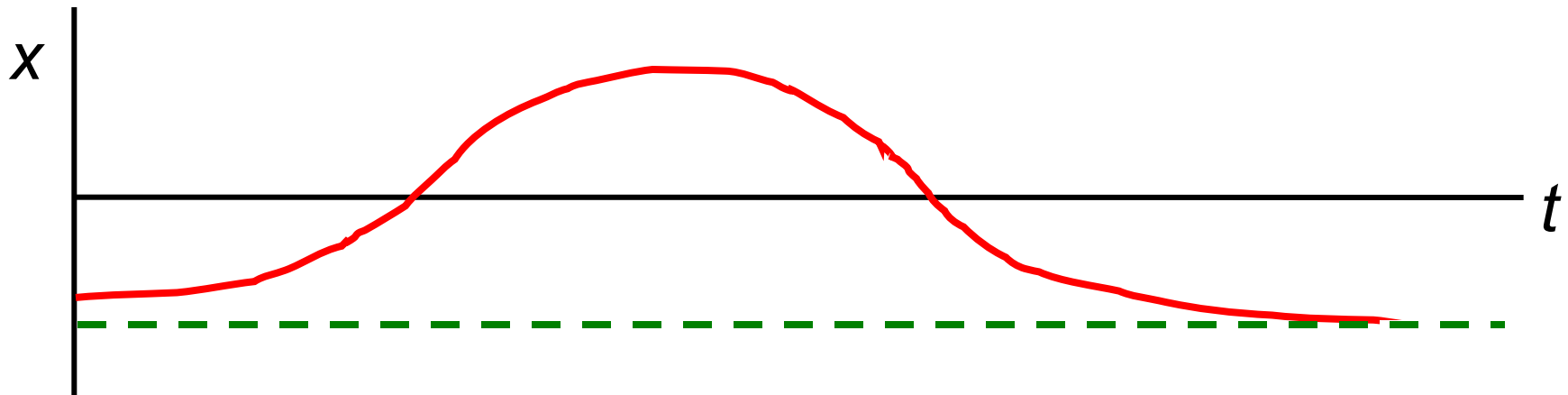
Area below the time axis = backward (negative) displacement.

"Net area" (above - below) = displacement.

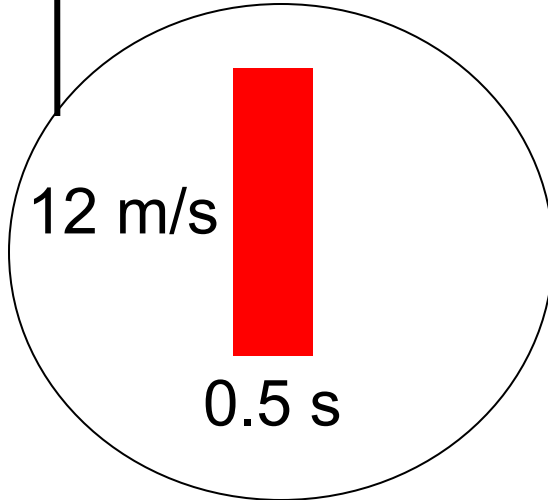
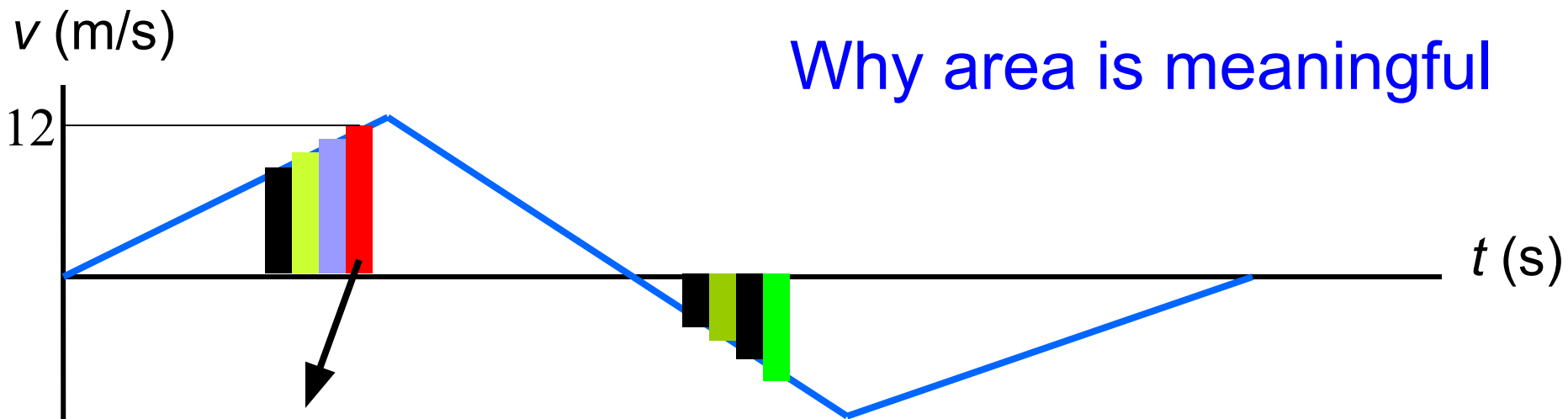
Total area (above + below) = total distance traveled.



The areas above and below are about equal, so even though a significant distance may have been covered, the displacement is about zero, meaning the stopping point was near the starting point. The position graph shows this too.



Why area is meaningful



- Imagine approximating the area under the curve with very thin rectangles.
- Each has area of height $\times$ width.
- The height is in m/s; width is in seconds.
- Therefore, area is in meters!
- The rectangles under the time axis have negative heights, corresponding to negative displacement.

Graphs of a ball thrown straight up

The ball is thrown from the ground, and it lands on a ledge.

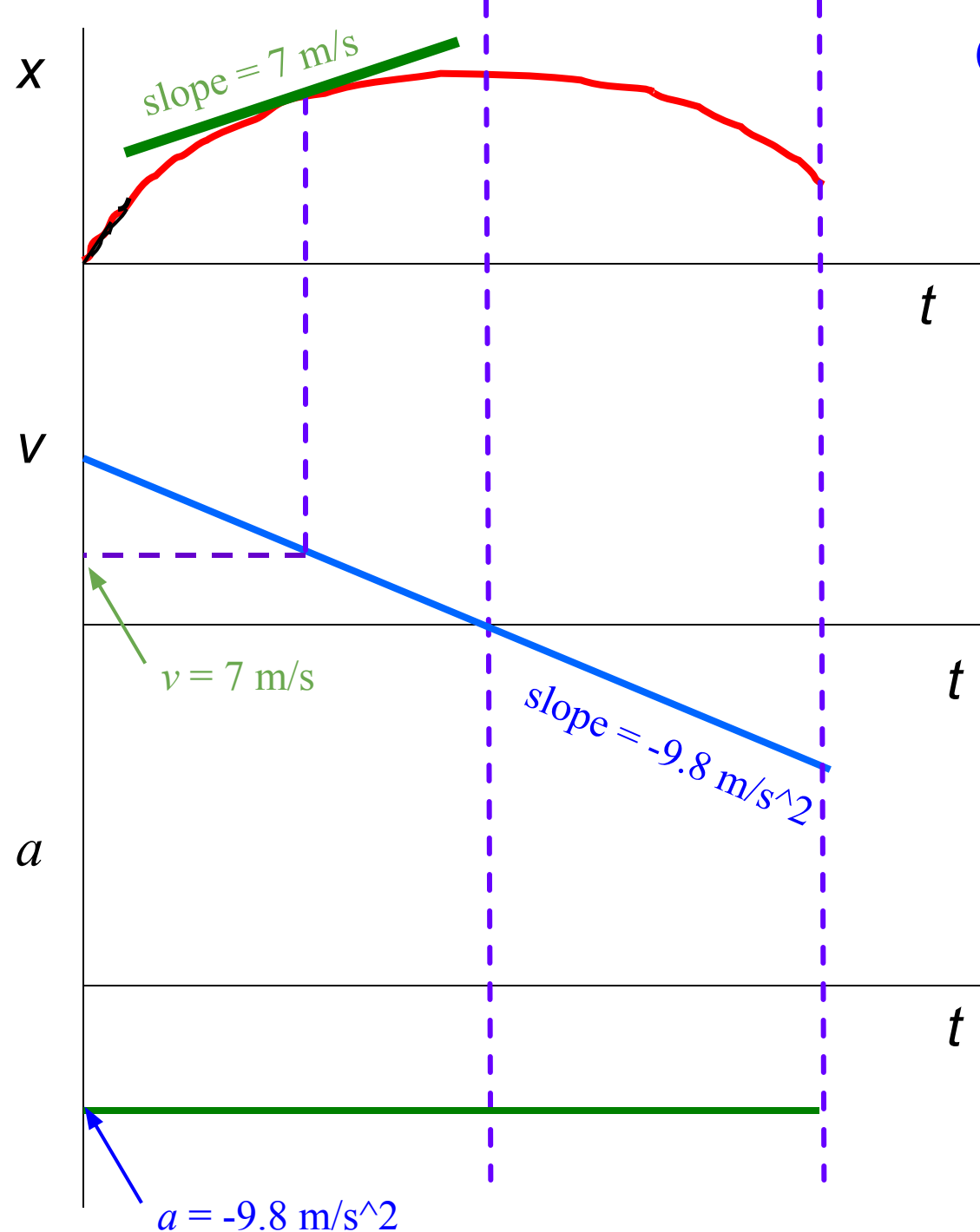
The position graph is parabolic, since acceleration is constant.

The ball peaks when the parabola is at its vertex.

The v graph has a slope of -9.8 m/s^2 (without air resistance).

Map out the slopes!

There is more “positive area” than negative on the v graph.



Graph Practice

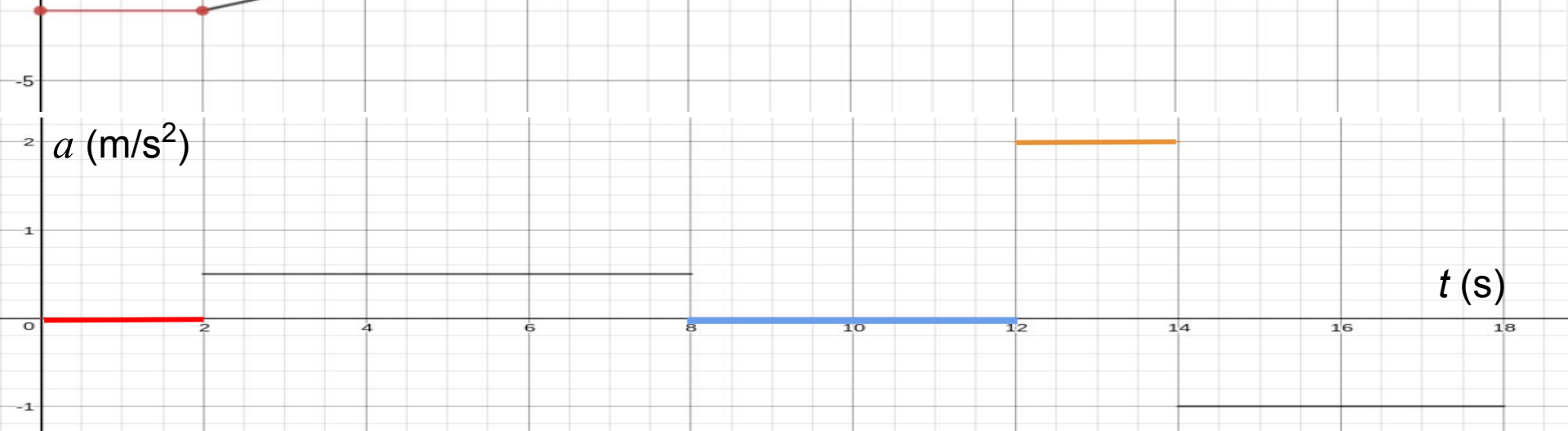
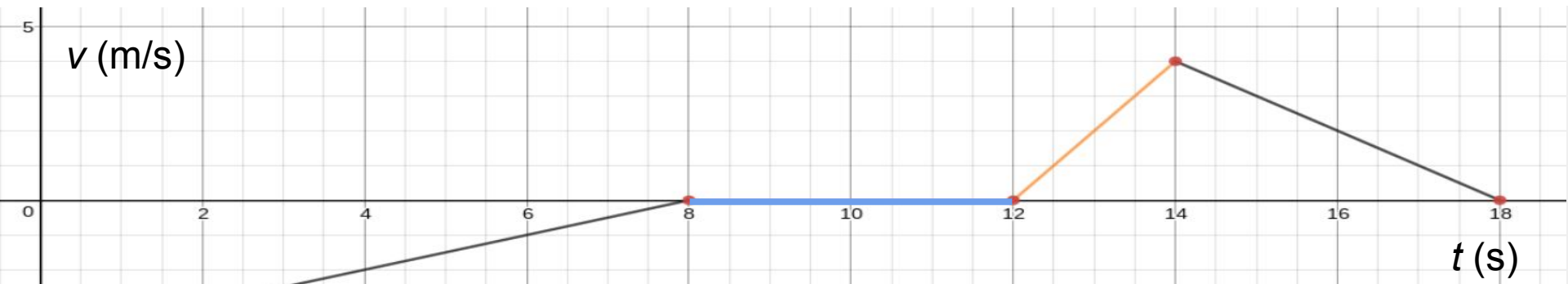
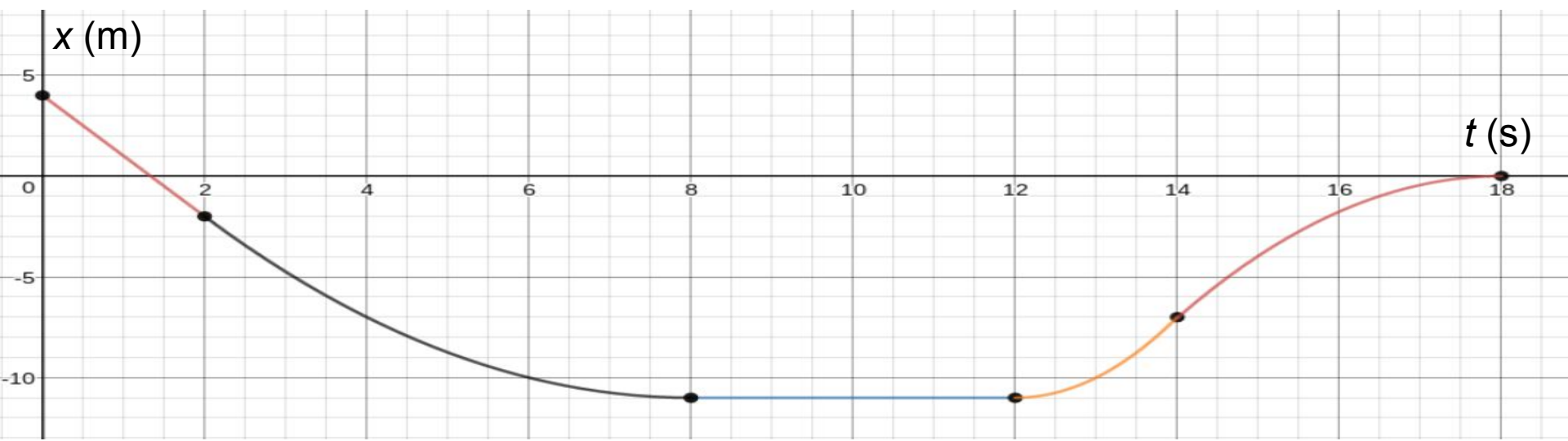
Try making all three graphs for the following scenario:

1. Schmedrick starts out north of home. At time zero he's driving a cement mixer south very fast at a constant speed.
2. He accidentally runs over an innocent moose crossing the road, so he slows to a stop to check on the poor moose.
3. He pauses for a while until he determines the moose is squashed flat and dead as a doornail.
4. Fleeing the scene of the crime, Schmedrick takes off again in the same direction, speeding up quickly.
5. When his conscience gets the better of him, he slows, turns around, and returns to the crash site.

Practice Quiz Question

Make all three graphs for the following scenario. Try it on your own first. Then compare your graphs with the answer on the next slide. Let east (E) be positive and “home” be our reference point.

- A knight in shining armor on a pogostick begins 4 m E of home already heading W at a constant speed until she is 2 m W of home.
- Then she continues W on her trusty steed for another 9 m as she uniformly slows to a stop over a period of 6 s.
- Now she bounces up and down in place there for 4 s.
- At this point she uniformly speeds up heading E, 2 s later reaching a max speed of 4 m/s after bouncing 4 m.
- Finally, as she approaches home, she slows uniformly to a stop over a period of 4 s.



Practice Quiz Question Follow-up

Answer these based on your graphs and the given info (answers on next slide).

1. At $t = 4.5$ s, is her acceleration positive or negative? Provide *three* graphical reasons.
2. What do the graphs show when velocity isn't changing?
3. At $t = 4$ s, what is the knight's position, velocity, and acceleration?
4. Explain how to calculate the acceleration at $t = 4$ s.
5. At $t = 4$ s, is she speeding up or slowing down? Explain fully.
6. At $t = 13$ s, in which direction is she heading? Give two reasons.
7. When is she at max speed? Give *two* reasons.
8. There is a tiny error at the right end of the v vs. t graph that could not be recognized were it not for the fact this scenario contains numerical values. What is the error? Explain.
9. What are her average speed and velocity over the whole trip?

Practice Quiz Question Follow-up Answers

1. Positive, because at this time: (1) the acceleration graph is obviously in positive territory (above the time axis); (2) the velocity graph has a positive slope; and (3) the position graph is concave up. By the way, the exact opposite is true in the last section of the graphs.
2. The a graph is zero, the v graph is flat, and the x graph is linear.
3. Reading off the graphs, $x = -7$ m (your graph might vary slightly), $v = -2$ m/s, and $a = +0.5$ m/s².
4. Find the slope of the v graph at that time:
$$(0 - (-3 \text{ m/s})) / 6 \text{ s} = 0.5 \text{ m/s}^2$$
5. Slowing down. The x graph shows the steepness (abs. val. of the slope) to be diminishing at this time. Also, the v graph shows the abs. val. of velocity (the speed) to be decreasing. Finally, v and a have opposite signs at this time. Any of these can be used to conclude that she's slowing down. *Cont. -->*

Practice Quiz Question Follow-up Answers (cont.)

6. East, simply because the value of the velocity at this time is positive. The slope of the v graph is irrelevant, but the slope of the x graph is positive, which also means she's going east. (The fact that x is negative at this time doesn't matter.)

7. According to the v graph, the max speed occurs at $t = 14$ s, where the graph is farthest from the time axis. You could also look at the x graph and see that it is steepest at this time.

8. The v graph shows the max velocity to be 4 m/s. However, this leads to an inconsistency. To see why, use the x graph to find the displacement: -4 m. This equals the "net area" under v vs. t graph. Under the time axis on the v graph (on the left) is an area of 15 m, meaning a -15 m displacement. Thus on the right end the area should be 11 m, since $-15 + 11 = -4$. But the area of that triangle is 12 m.

Continued -->

Practice Quiz Question Follow-up Answers (cont.)

9. To find the average speed, we find the total distance traveled and divide by the total travel time. From the x graph we see she traveled 15 m west, paused, and then traveled 11 m east, for a total of 26 m. So, her average speed was $26 \text{ m} / 18 \text{ s} = 1.44 \text{ m/s}$. Her average velocity is her displacement over the time: $-4 \text{ m} / 18 \text{ s} = 0.22 \text{ m/s}$.

Note: At any instant of time, an object's speed (a scalar) is the magnitude of its velocity (a vector), so in 1 D, they'll differ by at most a sign. However, as exemplified above, the average speed (a scalar) is not necessarily the magnitude of the average velocity (a vector). This is because the distance traveled (a scalar) is only the magnitude of the displacement (a vector) when the object's direction does not change.

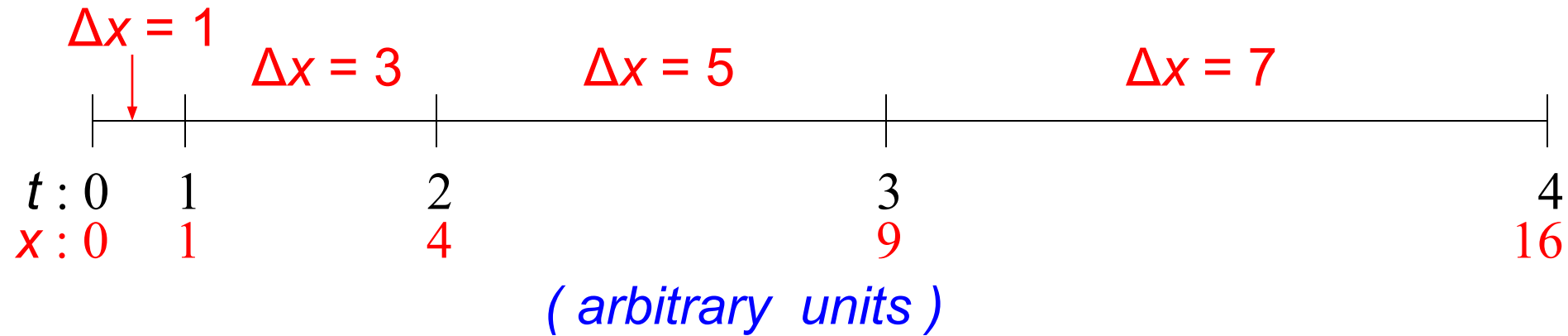
Kinematics Practice

A catcher catches a 90 mph fastball. His glove compresses 4.5 cm. How long does it take to come to a complete stop? Be mindful of your units!

2.24 ms

Answer

Uniform Acceleration



When an object starts from rest and undergoes constant acceleration:

- Position is proportional to the square of time.
- Changes in position (displacement) result in the sequence of odd numbers.
- Falling bodies exhibit this type of motion (since g is constant).

Relationships

Let's use the kinematics equations to answer these:

1. A mango is dropped from a height h .

- a. If dropped from a height of $2h$, would the impact speed double?
- b. Would the air time double when dropped from a height of $2h$?

2. A mango is thrown down at a speed v .

- a. If thrown down at $2v$ from the same height, would the impact speed double?
- b. Would it take half the time to hit the ground?

3. A rubber chicken is launched straight up at speed v from ground level. Find each of the following if the launch speed is tripled (in terms of any constants and v).

a. max height

$$9v^2/2g$$

b. hang time

$$6v/g$$

c. impact speed

$$3v$$

Answers